# Detecting ”Nine-Dash Line” Images in Digital Content via Faster R-CNN and DINOv2-Based Knowledge Distillation

***Abstract*—The “nine-dash line” image represents an illegal maritime claim that frequently appears in digital content, posing significant risks of distorting Vietnam’s territorial sovereignty. Manual moderation cannot keep pace with the rapidly growing volume of data, especially as such infringing images are often small, thin, and subtly embedded. In this study, we employ an enhanced Faster R-CNN model, incorporating knowledge distillation (KD) from the vision-language model DINOv2 to detect “nine-dash line” imagery. A KD loss function is utilized to align embeddings between predicted and ground-truth regions, thereby improving the detection of small and occluded objects. The model is trained on a specialized dataset of 836 images and achieves a mean Average Precision (mAP) of 58.4%, representing an improvement of 6.2% over the baseline Faster R-CNN with a ResNet-50 backbone. In addition, the employed approach achieves an Average Recall (AR) of 66.2%, compared to 58.8% for the baseline model. This system demonstrates strong potential for integration into automated digital content moderation workflows, supporting the protection of national sovereignty in cyberspace.**

***Index Terms*—Object detection, Knowledge distillation, Faster R-CNN**

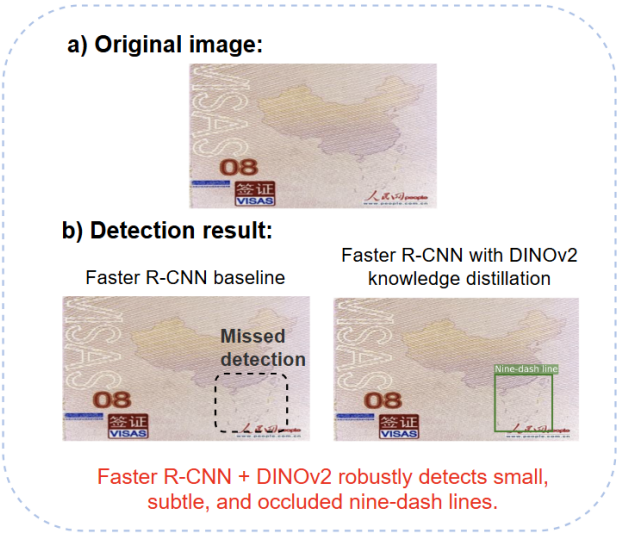


Fig. 1. Detection result comparison between baseline Faster R-CNN and Faster R-CNN with DINOv2 knowledge distillation on a challenging sample. The proposed method successfully detects the subtle nine-dash line, which is missed by the baseline.

## I. Introduction

Object Detection [1] is a fundamental task in the field of Computer Vision that involves identifying the presence, location, and category of one or more objects within an image or video frame. Unlike image classification, which assigns a single label to an entire image (e.g., “cat”), object detection precisely localizes each instance of an object using bounding boxes and assigns a class label to each (e.g., “cat” at coordinates x, y, with defined width and height). This enables computers to interpret and analyze image content in a way that more closely mimics human visual perception. It also facilitates downstream tasks such as object tracking across video frames, counting object instances, or triggering alerts based on detected objects. Object detection has broad applications across various domains, including autonomous driving, surveillance systems, robotics, and medical image analysis. The “Nine-Dash Line” is an illegal maritime claim unilaterally asserted by China in the South China Sea, which contradicts international law—specifically the 1982 United Nations Convention on the Law of the Sea (UNCLOS)—and was officially rejected by the Permanent Court of Arbitration (PCA) in 2016 [2]. However, depictions of the Nine-Dash Line continue to appear frequently in digital content such as maps, mobile applications, films, educational materials, and online media platforms. This poses significant risks of distorting Vietnam’s territorial sovereignty. In the context of globalization, digital products from diverse sources are increasingly entering the Vietnamese market, amplifying the risk of unlawful content dissemination. Meanwhile, content moderation efforts are currently performed primarily through manual inspection, which is insufficient to handle the massive volume of data [3]. Furthermore, infringing images are often subtly embedded as small, inconspicuous details, making them difficult to detect and often beyond the capabilities of traditional image search and detection tools.

This study employs a Faster R-CNN [4] [5] [6] model with a ResNet-50 backbone, integrating knowledge distillation (KD) [7] from the vision-language model DINOv2 [8]. Specifically, we apply a novel KD loss function that leverages the strong embeddings from DINOv2 to align predicted regions with ground-truth, aiming to enhance the detection of small and occluded objects. The model is trained on a specialized dataset and demonstrates an increase in mAP of 6.2% (from 52.2% to 58.4%) and an improvement in Average Recall (AR) from 58.8% to 66.2% compared to the baseline Faster R-CNN. These results indicate the practical potential of the proposed

approach for automated digital content moderation.

To provide a more intuitive illustration of the problem and our method's effectiveness, Figure 1 shows a challenging case in which the baseline Faster R-CNN fails to detect a subtle nine-dash line, while our method with DINOv2 knowledge distillation successfully localizes it. This example highlights the significance of incorporating vision-language knowledge for detecting small, subtle, or occluded objects in complex digital content.

## II. Related works

In recent years, convolutional neural network (CNN) models have made significant advances in object detection. The development began with R-CNN, which uses selective search to generate region proposals and CNNs to extract features [9], followed by Fast R-CNN, which improves processing speed through global feature computation and RoI Pooling [10]. Faster R-CNN further advances the field by replacing the manual region proposal step with a Region Proposal Network (RPN), enabling end-to-end training and enhancing overall efficiency. In parallel, one-stage methods such as YOLO [11] and SSD [12] approach the problem as regression tasks, significantly accelerating processing and making them suitable for real-time applications. However, these models often compromise on accuracy, particularly when detecting small or occluded objects. Recently, research has focused on leveraging knowledge from large pre-trained models based on the Vision Transformer (ViT) architecture [13]. ViT models are capable of learning semantically rich and highly generalizable embeddings, which effectively support a wide range of computer vision tasks. Integrating knowledge from ViT models through knowledge distillation opens up new directions for improving object detection performance, especially in challenging cases such as small or occluded objects.

Faster R-CNN is not only one of the pioneering models in the field but also widely applied due to its accuracy and effectiveness in detecting complex objects. Notable applications include medical image analysis (e.g., detecting and classifying cells in brightfield microscopy images of malaria-infected blood [14] ), intelligent transportation systems (identifying traffic signs, vehicles, and pedestrians) [15] , and automatic license plate recognition [16]. In these domains, the model not only delivers accurate detection but also adapts well to diverse environmental conditions. However, most current applications focus on objects with stable shapes, high-quality images, sufficiently large sizes, and minimal occlusion. Notably, there is still limited research applying CNNs to sensitive digital content moderation tasks related to national sovereignty protection—where the target objects are often small, thin, variable in shape, and subtly embedded. Therefore, this study aims to leverage the deep learning capabilities of CNNs to automatically detect images containing violations in digital documents, serving the needs of sensitive content moderation in Vietnam.

## III. Method

We apply a two-branch approach: (1) object detection using the Faster R-CNN model, and (2) knowledge distillation from a vision-language model [17]. Figure 2 provides an overview of the methodology we implement. In the knowledge distillation branch, we use a vision-language encoder (V-L Encoder) to extract embeddings from both the predicted object regions and the corresponding ground-truth regions. Knowledge distillation techniques are then employed to align the predicted embeddings with the ground-truth embeddings. The training process is optimized by combining standard loss functions with a knowledge distillation-based loss, thereby improving the model's accuracy. The technical components applied include the Faster R-CNN model with a CNN backbone, Region Proposal Network (RPN), and ROI Pooling to generate predicted object regions and convert them into fixed-size representations.

### A. Faster R-CNN

The basic training procedure of Faster R-CNN for generating predicted object regions consists of three main steps: (1) input image preprocessing, (2) the object detection stages of Faster R-CNN, and (3) the loss functions used during training.

*1) Image Preprocessing:* Input images are normalized for each color channel by subtracting the mean and dividing by the standard deviation. Next, the dataset is augmented using techniques such as image rotation, blurring, adjustments of brightness and color, and aspect ratio changes to increase diversity and enhance the model's generalization capability across various input conditions [18]. After preprocessing and augmentation, images are grouped into batches according to the available memory and computational resources.

*2) Object Detection Stages of Faster R-CNN:* **CNN Backbone:** After preprocessing, input images are passed through a CNN backbone (such as ResNet-50) that acts as a feature extractor. The CNN consists of multiple convolutional and nonlinear layers, allowing the model to learn hierarchical representations from the original images. The early layers focus on detecting basic features like edges, corners, or simple textures, while deeper layers capture more complex and abstract features such as object shapes or semantic structures. The output of the backbone is a multi-dimensional feature map, rich in spatial and semantic information, which serves as input for the subsequent stages.

**Region Proposal Network (RPN):** The feature map from the backbone is fed into the Region Proposal Network (RPN)—a lightweight convolutional network responsible for identifying potential Regions of Interest (ROIs). The RPN slides a small window across the feature map, generating multiple anchor boxes of different scales and aspect ratios at each position. Each anchor box is evaluated through two branches: a classification branch predicting objectness (i.e., the likelihood of containing an object) and a regression branch refining the anchor box's position and size to better match the object. The RPN is trained using both binary classification loss and regression loss to optimize both branches. The output is

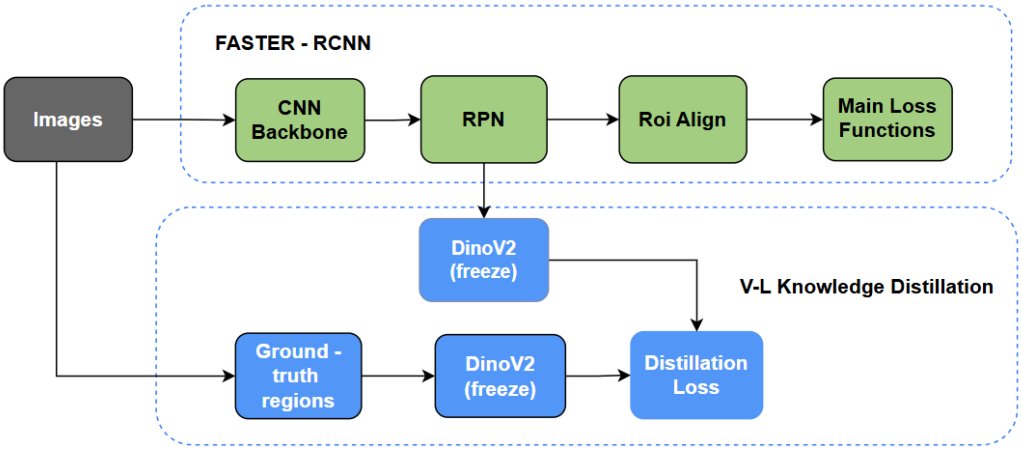


Fig. 2. Overview of the applied method.

a set of proposed object regions along with their confidence scores, representing areas with high likelihood of containing objects.

**ROI Pooling:** The proposed regions (ROIs) from the RPN have varying sizes, making further processing challenging. To address this, the ROI Pooling layer converts each ROI into a fixed-size feature map (e.g., 7×7 or 14×14 pixels) [10]. This is achieved by dividing each ROI into a fixed number of spatial bins and applying pooling (usually max pooling) within each bin, thereby preserving essential information. This approach enables the model to process diverse ROIs consistently, improving the efficiency and stability of subsequent classification and regression steps.

**Object Classification and Bounding Box Regression:**The fixed-size feature maps from ROI Pooling are fed into a series of fully connected (FC) layers to perform two tasks: object classification and bounding box regression. The classification branch determines the object class (or background) for each ROI, while the regression branch refines the bounding box coordinates to best fit the actual object in the image. The final output consists of a set of predictions, each comprising a class label and an accurate bounding box, representing the objects detected in the input image.

*3) Main Loss Functions:* Faster R-CNN is trained by jointly optimizing multiple loss components [4] [5] [6], each corresponding to a key stage in the model:

- RPN Classification Loss ($L_{rpncls}$): Binary cross-entropy loss measures the accuracy of classifying anchor boxes as object or background, helping the RPN generate high-quality proposals.
- RPN Regression Loss ($L_{rpnr}$): Smooth L1 loss calculates the error between predicted and ground-truth anchor box coordinates, enabling the RPN to refine proposal positions and sizes.
- ROI Classification Loss ($L_{roic}$): Multi-class cross-entropy loss classifies objects within each ROI after RoI Pooling, ensuring accurate class predictions.
- ROI Regression Loss ($L_{roir}$): Smooth L1 loss further adjusts the bounding box coordinates for each ROI, improving localization accuracy.

### B. Vision-language model knowledge distillation

This section details the methodology we apply, in which knowledge from a pre-trained vision-language (V-L) model is distilled to improve the performance of the Faster R-CNN object detection model. Three main aspects are addressed: (1) extracting vision-language embeddings from object regions, (2) constructing a loss function based on knowledge distillation, and (3) integrating this with the main loss functions of the model.

*1) Extracting V-L embeddings from object regions:* **Extracting V-L embeddings from ground-truth object regions:** For our dataset, manually annotated bounding boxes are used to determine the position and size of objects. These regions are passed through a pre-trained vision-language encoder (V-L Encoder), which transforms the visual information into embeddings that incorporate knowledge from both the image and language domains. **Extracting V-L embeddings from predicted object regions:** Features from the proposed regions (the output of ROI Align) are also passed through the same V-L encoder to generate the corresponding V-L embeddings. **Embedding normalization:** To ensure consistency and improve compatibility between embeddings when computing the loss, we apply L2 normalization, which enforces uniform vector length and facilitates comparison and model optimization.

*2) Loss functions based on knowledge distillation techniques:* In this study, we apply two loss functions based on knowledge distillation to align the predicted embeddings from Faster R-CNN with those of the ground-truth object regions, thereby improving detection accuracy. Specifically, the first loss function matches each predicted embedding with

its corresponding ground-truth embedding, reducing the distance between these representations. **L1**: Manhattan distance between the predicted embedding ($\mathbf{e}_i^{\text{pred}}$) and ground-truth ($\mathbf{e}_i^{\text{gt}}$):

$$L_1 = \frac{1}{K}\sum_{i=1}^{K} \|\mathbf{e}_i^{\text{pred}} - \mathbf{e}_i^{\text{gt}}\|_1$$

The second loss function focuses on preserving the relationships among all embeddings within the set of object regions, ensuring that the correlation structure between the predicted and ground-truth embeddings is maintained. **IRM**:Frobenius norm of the difference between the correlation matrices of the predicted embeddings ($\mathbf{S}_{\text{pred}}$) and ground-truth ($\mathbf{S}_{\text{gt}}$):

$$L_{\text{irm}} = \frac{1}{K^2}\|\mathbf{S}_R - \mathbf{S}_I\|_F^2$$

where $\mathbf{S}_R$ is a $K \times K$ matrix of V-L embedding correlations from predicted object regions by Faster R-CNN, and $\mathbf{S}_I$ is a $K \times K$ matrix of V-L embedding correlations from ground-truth object regions. $\|\mathbf{S}_R - \mathbf{S}_I\|_F^2$ denotes the squared Frobenius norm of the difference between the two matrices. The total knowledge distillation loss is computed as follows:

$$L_{KD} = \beta_1 \cdot L_1 + \beta_2 \cdot L_{\text{irm}}$$

The final total loss used to train the model combines these knowledge distillation losses with the main Faster R-CNN loss terms: The total training loss combines the main losses of Faster R-CNN and the KD loss:

$$L_{\text{Total}} = L_{\text{rpnc}} + L_{\text{rpnr}} + L_{\text{roic}} + L_{\text{roir}} + L_{KD}$$

## IV. Experiment

### A. Dataset

In this study, we construct a specialized dataset specifically designed for the task of detecting the "nine-dash line" imagery in digital publications. The dataset comprises a total of 836 images, collected from a variety of sources such as maps, software interfaces, screenshots from websites, digital documents, and other digital products that may contain sovereignty-violating visual elements. The data construction process involves manual annotation , where each image is labeled with a bounding box indicating the location of the "nine-dash line," along with the corresponding class label. The dataset is split into training and validation sets with a 90/10 ratio, resulting in 752 images for training and 84 images for validation. This proportion is chosen to ensure that the model has sufficient data to learn diverse features while retaining a separate subset for evaluating overall performance. Although smaller in size compared to standard benchmark datasets, our custom-built dataset is highly domain-specific and focused, making it well-suited for detecting small, irregularly shaped, and potentially occluded objects—typical characteristics of "nine-dash line" imagery subtly embedded in digital content. To further illustrate the characteristics of our dataset, we present a statistical analysis of image and object size distributions. Figure 3 shows the distribution of image widths and heights in the dataset,

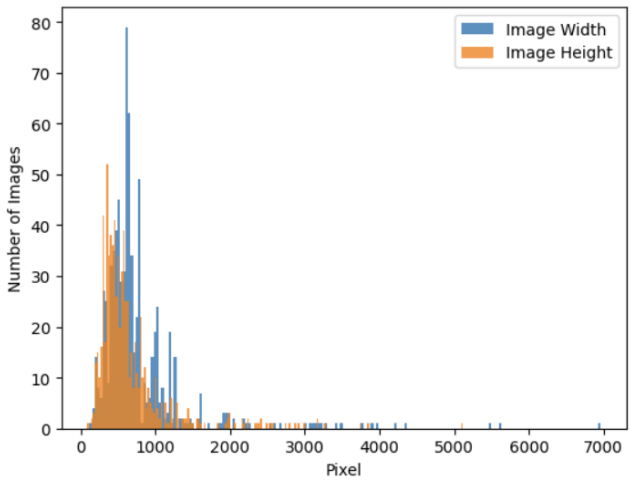


Fig. 3. Histogram of image width and height distributions in the nine-dash line dataset.

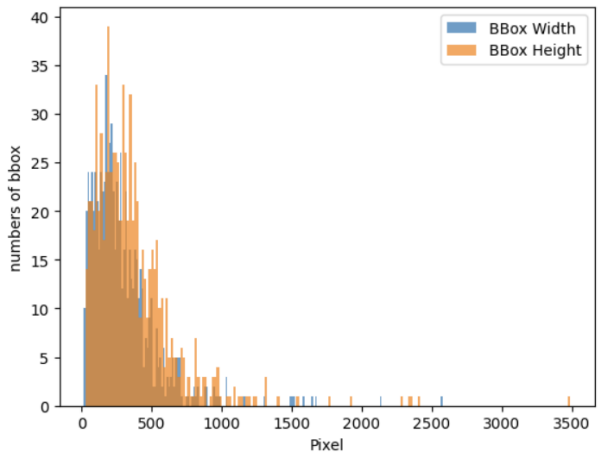


Fig. 4. Distribution of bounding box width and height for annotated "nine-dash line" regions.

indicating a wide range of resolutions, with most images falling below 1000 pixels in both dimensions. This diversity in image sizes poses challenges for object detection models, particularly in handling objects at varying scales.

In addition, we examine the distribution of bounding box (bbox) sizes, as shown in Figure 4. Most bounding boxes representing the "nine-dash line" are relatively small, with both widths and heights typically below 500 pixels. This reinforces the fact that the task requires robust detection of small, thin, and irregularly shaped objects, further motivating the use of advanced detection techniques.

### B. Implementation Details

*1) Data Preparation:* The input data preparation process is a crucial step, playing a key role in ensuring the accuracy and training efficiency of the Faster R-CNN model. Prior to being fed into the model, the images undergo pixel value normalization by subtracting the mean and dividing by the

standard deviation for each color channel. Specifically, each pixel is standardized to have a mean of zero and a standard deviation of one, bringing the entire dataset into a consistent distribution. This step not only reduces the impact of lighting variations among images but also facilitates faster and more stable model convergence during training. Additionally, to increase data diversity and reduce the risk of overfitting, data augmentation techniques are applied. Several methods used include image rotation, brightness and contrast adjustment, as well as mild blurring or addition of Gaussian noise. These operations enable the model to learn more generalized features, thereby improving its ability to detect "nine-dash line" images under varying presentation conditions and image qualities.

*2) Training Procedure:* The training process is implemented in a well-structured manner, consisting of data preparation, model initialization, training, and evaluation. All hyperparameters are carefully configured to optimize learning efficiency. The initial learning rate is set to 0.005 and is reduced by a factor of 10 every 5 epochs, allowing the model to converge stably. The batch size is set to 4 to balance training speed, memory usage, and generalization capability. Stochastic Gradient Descent (SGD) is used as the optimizer, with a weight decay of 0.0005 and a momentum of 0.9, to improve convergence and reduce overfitting. The entire training process lasts for 20 epochs, which is sufficient for the model to learn important features from the data.

In terms of architecture, we employ Faster R-CNN with a ResNet-50 backbone, a popular and effective choice for object detection tasks. To extract visual-language (V-L) embeddings from both predicted and ground-truth regions, we integrate a pre-trained DINOv2 encoder into the knowledge distillation framework. The hyperparameters $\beta_1$ and $\beta_2$ are set to 1 and 0.1 in our experiments, playing a crucial role in balancing the contributions of different loss components during training.

### C. Evaluation Metrics

Mean Average Precision (mAP) is one of the standard and most important metrics in object detection tasks. mAP measures the average accuracy of a model across all object classes by calculating the area under the Precision-Recall curve (AUC) for each class and then averaging these values. This metric reflects the model's ability to balance precision—the proportion of correct predictions among all predictions—and recall—the proportion of actual objects detected. In this report, we use mAP, a stringent variant commonly applied to the COCO dataset, to evaluate model performance across a range of IoU thresholds.

Average Recall (AR) is another evaluation metric that focuses on the model's ability to detect all objects present in an image. AR measures the average proportion of actual objects that are correctly detected by the model, averaged over all classes and IoU thresholds. This metric is particularly important in applications requiring high sensitivity, where missing objects could lead to serious consequences.

### D. Result

*1) main result:* The experimental results, as presented in Table I, provide a comprehensive comparison between baseline Faster R-CNN models and the improved variant incorporating DINOv2 knowledge distillation, as well as other state-of-the-art object detection architectures such as DETR and SSD. The findings demonstrate that integrating knowledge distillation from DINOv2 with a ResNet-50 backbone yields the highest detection performance, achieving a mAP of 58.4% and an Average Recall (AR) of 66.2%. In contrast, the baseline Faster R-CNN with ResNet-50 achieves a mAP of 52.2% and AR of 58.8%, while ResNet-101 provides a mAP of 50.6% and AR of 56.4%. This clearly shows that the proposed knowledge distillation approach with DINOv2 not only outperforms both ResNet-50 and ResNet-101 baselines, but also demonstrates better generalization for challenging detection tasks.

Comparison with other popular object detection models, such as DETR and SSD, further reinforces the advantages of the proposed approach. DETR with ResNet-50 achieves a mAP of only 32.7% and an AR of 49.2%, while SSD with VGG16 yields the lowest performance (mAP 31.0%, AR 42.6%). These results highlight that the improved Faster R-CNN with knowledge distillation from DINOv2 significantly outperforms these baselines in detecting small, irregularly shaped, and occluded objects—typical characteristics of "nine-dash line" imagery. In particular, the low performance of DETR can be explained by the fact that most bounding boxes in this dataset are extremely small, with many cases measuring less than 20×20 pixels after preprocessing with the DetrImageProcessor. When bounding boxes are this small, transformer-based models like DETR may struggle to preserve and distinguish important features, as spatial information can be lost or mixed during the attention layers. This results in inferior detection of small, thin, or occluded objects compared to the improved Faster R-CNN approach.

Overall, the consistent improvement in both mAP and AR underscores the value of leveraging advanced visual-language representations and knowledge distillation for automated content moderation tasks. The proposed method not only outperforms the traditional baselines but also demonstrates strong potential for practical deployment in real-world scenarios that demand high accuracy and recall in sensitive object detection applications.

TABLE I
PERFORMANCE COMPARISON BETWEEN BASELINE FASTER R-CNN AND FASTER R-CNN WITH DINOV2 KNOWLEDGE DISTILLATION.

| **Model** | **CNN Backbone** | **mAP** | **AR** |
|---|---|---|---|
| Faster R-CNN with knowledge distillation | ResNet-50 | 58.4% | 66.2% |
| Faster -RCNN | ResNet-50 | 52.2% | 58.8% |
| Faster -RCNN | ResNet-101 | 50.6% | 56.4% |
| DETR | ResNet-50 | 32.7% | 49.2% |
| SSD | VGG16 | 31.0% | 42.6% |

*2) The importance of the hyperparameters $\beta_1$ and $\beta_2$:* The selection of the hyperparameters $\beta_1$ and $\beta_2$ plays a crucial

role in determining the model's performance. The results in Table II clearly demonstrate the impact of varying these values. For example, setting $\beta_1 = 1.0$ and $\beta_2 = 0.1$ yields the highest mAP of 58.4%. When $\beta_2$ is increased to 0.5 (with $\beta_1 = 1.0$), the mAP decreases to 53.9%. Similarly, with $\beta_1 = 0.5$ and $\beta_2$ increasing from 0.1 to 0.5, the mAP drops from 54.9% to 54.1%. Additionally, the configuration $\beta_1 = 1.0$ and $\beta_2 = 1.0$ results in a mAP of 54.8%, which is still lower than the best setting. Notably, with $\beta_1 = 0$ and $\beta_2 = 1.0$, the mAP reaches 55.4%, indicating that the loss component associated with $\beta_2$ also plays an important role.

These observations indicate that careful tuning of $\beta_1$ and $\beta_2$ is essential to optimize detection performance. In this study, the optimal configuration is $\beta_1 = 1.0$ and $\beta_2 = 0.1$, achieving the best result.

TABLE II
IMPACT OF $\beta_1$ AND $\beta_2$ VALUES ON MAP.

| $\beta_1$ | $\beta_2$ | mAP (%) |
|---|---|---|
| 1.0 | 0.1 | 58.4 |
| 1.0 | 0.5 | 53.9 |
| 0.5 | 0.1 | 54.9 |
| 0.5 | 0.5 | 54.1 |
| 1.0 | 1.0 | 54.8 |
| 0 | 1.0 | 55.4 |

*3) The importance of the CNN backbone:* In addition to optimizing loss parameters, the choice of CNN backbone architecture also has a significant impact on the model's final performance. As shown in Table III, Faster R-CNN with knowledge distillation and a ResNet-50 backbone achieves a mAP of 58.4% and an AR of 66.2%, substantially outperforming the ResNet-101 variant, which only reaches a mAP of 43.7% and AR of 53.7%. This result contrasts with the common assumption that deeper networks like ResNet-101 always deliver better results. On our specialized dataset, ResNet-50 proves more effective, likely because it provides a better balance between model complexity and generalization capability, and it is particularly well-suited to the new knowledge distillation approach. These findings highlight that selecting an appropriate backbone is a crucial factor, especially for tasks involving the detection of small, thin, and morphologically diverse objects such as the "nine-dash line" imagery.

TABLE III
DETECTION PERFORMANCE OF FASTER R-CNN WITH KNOWLEDGE DISTILLATION FOR DIFFERENT CNN BACKBONES.

| CNN Model | mAP (%) | AR (%) |
|---|---|---|
| ResNet 50 | 58.4 | 66.2 |
| ResNet 101 | 43.7 | 53.7 |

## V. CONCLUSION

In conclusion, this paper presents the application of an improved Faster R-CNN model enhanced with knowledge distillation from DINOv2 for detecting "nine-dash line" images in digital content. The proposed system achieves a mAP of 58.4%, marking an improvement of 6.2% over the baseline Faster R-CNN, primarily due to the integration of a knowledge distillation loss that effectively aligns predicted and ground-truth embeddings. These results highlight the potential of the approach for automated digital content moderation, contributing to efforts in safeguarding national sovereignty. For future work, we plan to expand the dataset and explore alternative detection architectures such as YOLOv9 or Vision Transformer (ViT) models to further enhance processing speed and robustness. .

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